

UFUG 2106 : Discrete Mathematics

Lecture 0: Course introduction



香港科技大学(广州)
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TECHNOLOGY (GUANGZHOU)

数据科学与分析学域
DATA SCIENCE AND ANALYTICS THRUST
信息枢纽
INFORMATION HUB

1. Existential Questions & Logistics

Who is this guy?



Course team

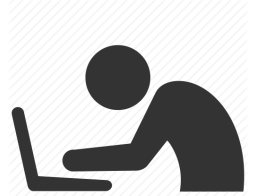
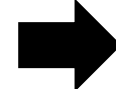
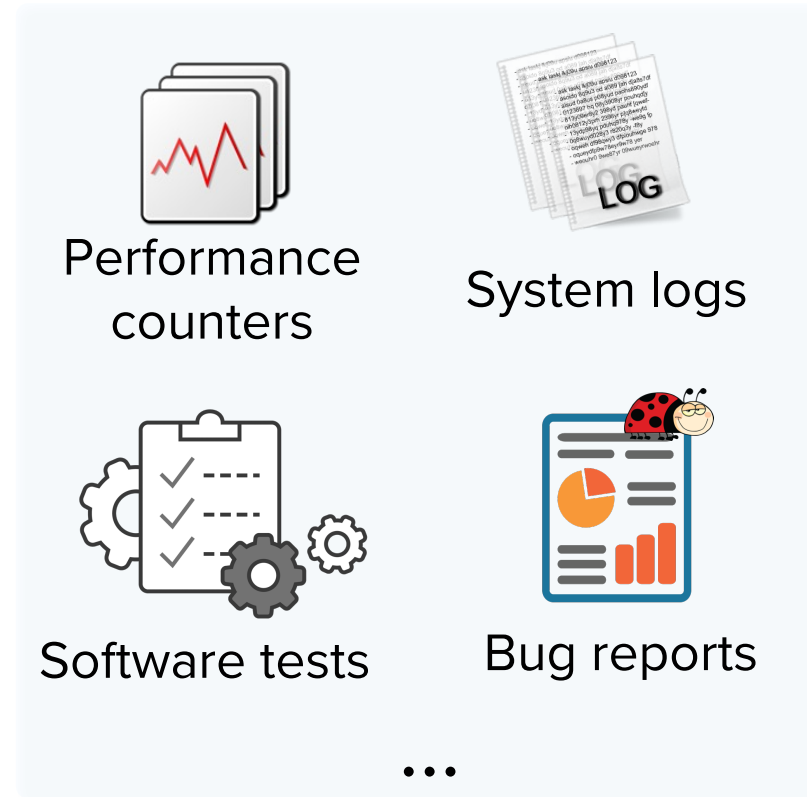
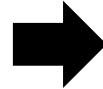


Zishuo DING

Ph.D. in ECE, 2024
University of Waterloo



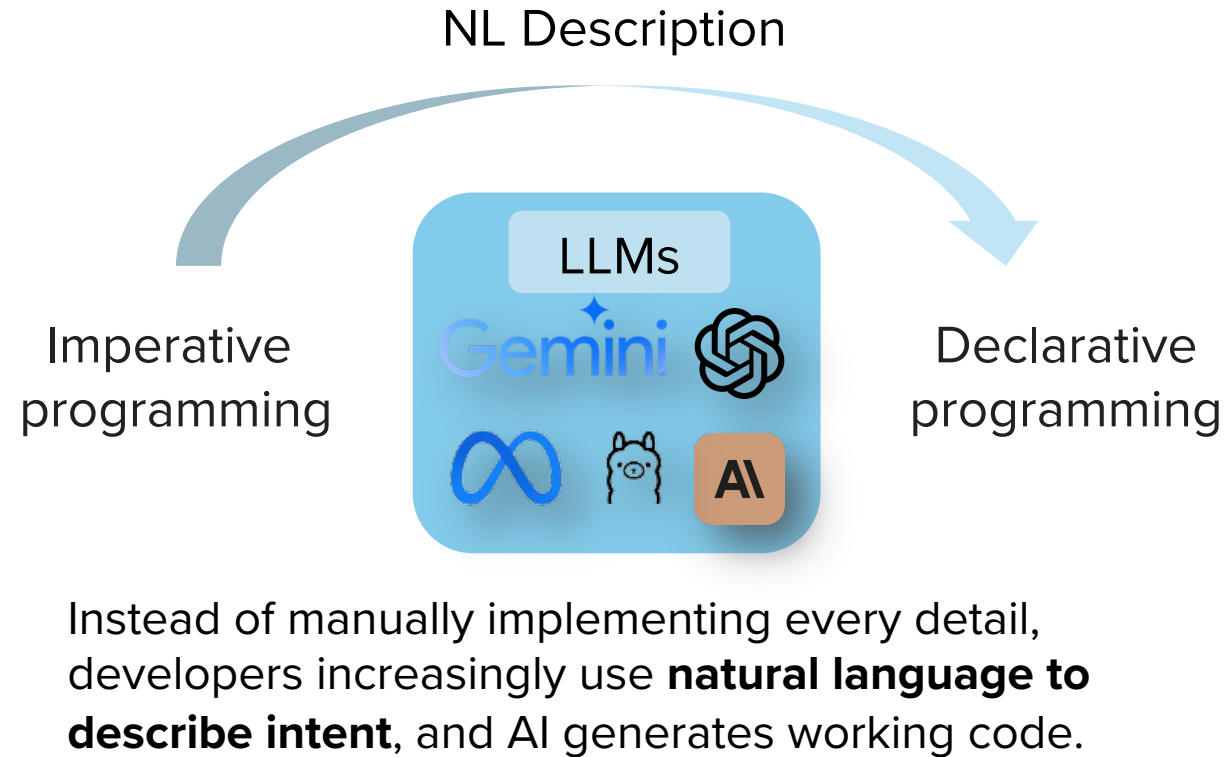
Software System



Software developers

Applying artificial intelligence, data analytics,
and static code analysis to provide automated
support to developers

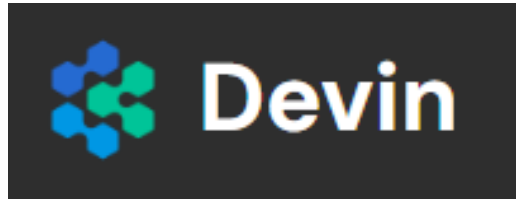
AI is transforming software development



AI is becoming the new programming interface for building software.

The rise of AI-powered coding

AI-powered coding assistants like Copilot, Cursor, and Amazon Q are reshaping how developers build software.



Leading AI coding copilots



The market is projected to reach **USD 2.8 billion by 2030.**

AI coding copilots are becoming a standard part of the modern development workflow.

Why software engineering perspective matters?

In today's world, AI systems are not just models, they are **complex software products** that must be engineered, deployed, evaluated, secured, and continuously improved.

← Post



Greg Brockman  
@gdb

With some exceptions, the biggest impacts in AI come from people who are experts at both software and machine learning.

Though most people expect the opposite, it's generally much faster to learn ML than software.

So great software engineers tend to have outsize potential in AI

The biggest breakthroughs in AI come from people who are experts in both software and machine learning.

— Greg Brockman, President & Co-Founder @ OpenAI

Course team



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**What is discrete
mathematics?**



Discrete Mathematics

- Discrete mathematics is a branch of mathematics that is concerned with “**discrete**” mathematical structures instead of “**continuous**”
- Continuous mathematics:
 - Real numbers
 - Calculus
 - Euclidian geometry
 - Etc.
- Discrete mathematics:
 - Integers, number theory
 - Logic
 - Set theory
 - Combinatorics
 - Graph theory
 - Etc.

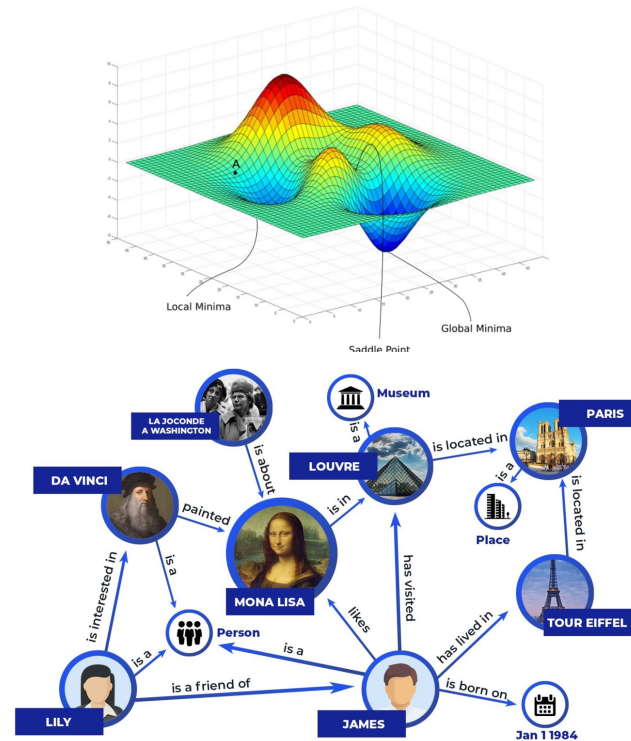


Image source: <https://community.atlassian.com/t5/Confluence-questions/Knowledge-graph/qaq-p/1565284>

Why does it matter?

- Digital computers are built as a discrete machine: 0 and 1
 - How to use 0s and 1s to solve real-world problems?
- A mathematical foundation for
 - Computer science topics: Data structures, algorithms, relational database, automata theory, formal languages, compilers, cryptography, artificial intelligence, etc.
 - Mathematics topics: linear algebra, abstract algebra, combinatorics, probability, logic and set theory, number theory
- Core knowledge in computer science curricular
 - Recommended by both IEEE (Institute of Electrical and Electronics Engineers) and ACM (Association for Computing Machinery)

**What do I do to
survive?**



What you are expected to learn (Intended Learning Outcomes)

By the end of this course, students should be able to:

1. Understand the formulation of common problems in several areas of discrete mathematics, including logic and proof, combinatorics, recursion and recurrences, and graph theory
2. Understand and analyze the key algorithms in the above areas of discrete mathematics
3. Plan and select appropriate concepts and algorithms in the above areas to solve real-world problems

What you'll be doing every week

This is *by far* the most important survival information.

- Lectures L02:

- Friday: 12:00PM - 02:50PM, Rm 149, E1

- Tutorials T02:

- Friday: 06:30PM - 07:20PM, Rm 147, E1

- Mid-term test: around the 7th week (tentative)

What topics are coming (Course Schedule (tentative))

Week	Topics	Weekly ILOs
1	Speaking mathematically	ILO1
2	The logic of compound statements	ILO1, ILO2, ILO3
3	The logic of quantified statements	ILO1, ILO2, ILO3
4	Number theory and methods of proof	ILO1, ILO2, ILO3
5	Sequences, mathematical induction, and recursion	ILO1, ILO2, ILO3
6	Set theory	ILO1, ILO2, ILO3
7	Mid-term test; Properties of functions	ILO1, ILO2, ILO3
8	Properties of relations	ILO1, ILO2, ILO3
9	Counting and probability	ILO1, ILO2, ILO3
10	Theory of graphs and trees	ILO1, ILO2, ILO3
11	Analysis of algorithm efficiency	ILO1, ILO2, ILO3
12	Regular expressions and finite-state automata	ILO1, ILO2, ILO3
13	Finite-state automata; course review	ILO1, ILO2, ILO3

Recommended textbooks

- Main textbook:

- “Discrete Mathematics with Applications”, 5th Edition, Susanna S. Epp
- With detailed explanations and lots of examples

- Extra topics

- More challenging, not covered by the textbook

- If you need more questions to practice, check another reference book:

- “Discrete Mathematics and Its Applications”, 8th Edition, Kenneth H. Rosen

Prerequisites

- Knowledge of CS principles and skills
- Familiar with (preferably object-oriented) programming
 - There will be a **coding tasks** to help you practice the knowledge you have learned.

How will I be evaluated?



Overall evaluation scheme

- Continuous Assessment: 30%
 - Written assignments (10%): around one assignment per Chapter
 - Course participation (5%): lectures, tutorials, quizzes
 - Projects (15%): one project
- Mid-term test: 25%
 - Closed book
- Final exam: 45%
 - Closed book

Written Assignments Rubric

Criteria	A (Excellent)	B (Good)	C (Satisfactory)	D (Marginal Pass)	F (Fail)
Understanding of Concepts	Thorough and deep understanding, no errors.	Good understanding with minor errors.	Basic understanding with some errors.	Limited understanding with significant errors.	Lacks understanding.
Problem Solving and Analysis	Correct, well-reasoned solutions, demonstrates creativity.	Mostly correct, lacks depth or creativity.	Partially correct with several errors.	Mostly incorrect with major errors.	Incorrect or missing solutions.
Clarity and Organization	Clear, logical, well-organized presentation.	Mostly clear, minor organizational issues.	Somewhat clear, lacks logical flow.	Disorganized, difficult to follow.	Lacks clarity and organization.
Mathematical Language and Notation	Correct and precise use of language and notation.	Minor errors in language or notation.	Some errors in language and notation.	Frequent errors in language and notation.	Incorrect use of language/notation.

Course Participation and Quizzes Rubric

Criteria	A (Excellent)	B (Good)	C (Satisfactory)	D (Marginal Pass)	F (Fail)
Engagement and Contribution	Actively participates, contributes meaningful insights.	Regular participation, relevant contributions.	Occasional participation, limited contributions.	Rarely participates or contributes.	Does not participate.
Accuracy	Answers accurate, demonstrate full comprehension.	Mostly accurate, few errors.	Some correct answers, significant errors.	Few correct answers.	Mostly incorrect or missing.
Completion	All questions attempted, thorough effort.	Most questions attempted, reasonable effort.	Some questions attempted.	Few questions attempted.	Little to no attempt.

Projects Rubric

Criteria	A (Excellent)	B (Good)	C (Satisfactory)	D (Marginal Pass)	F (Fail)
Research and Understanding	Comprehensive research, deep understanding.	Good research, clear understanding.	Basic research, some understanding.	Limited research and understanding.	Insufficient research/understanding .
Application of Concepts	Excellent application, demonstrates innovation.	Good application, some insight.	Basic application, limited insight.	Poor application of concepts.	Incorrect or no application.
Presentation and Organization	Clear, well-organized, engaging presentation.	Clear and organized presentation.	Somewhat organized, lacks clarity.	Disorganized and unclear.	Poor presentation.
Teamwork	Excellent collaboration, balanced contributions.	Good collaboration, mostly balanced.	Some collaboration, uneven contributions.	Poor collaboration.	Little to no collaboration.

Midterm and Final Examinations Rubric

Criteria	A (Excellent)	B (Good)	C (Satisfactory)	D (Marginal Pass)	F (Fail)
Understanding and Application	Excellent understanding and application of concepts.	Good understanding, minor errors.	Satisfactory understanding, some errors.	Limited understanding, major errors.	Insufficient understanding.
Problem Solving	Correct, thorough solutions, demonstrates insight.	Mostly correct solutions, minor gaps.	Partially correct solutions, significant gaps.	Mostly incorrect solutions.	Incorrect or missing solutions.
Clarity and Organization	Clear and well-organized responses.	Mostly clear and organized.	Somewhat clear, disorganized.	Poorly organized and unclear.	Disorganized and unclear.
Mathematical Language and Notation	Precise, correct use of language and notation.	Minor errors in language or notation.	Several errors in language and notation.	Frequent errors.	Incorrect use of language/notation.

Final Grade Descriptors

改成85-100的A category

Grades	Short Description	Elaboration on subject grading description
A	Excellent Performance	Demonstrates a comprehensive grasp of discrete mathematics, expertise in problem-solving, and significant creativity in mathematical reasoning and proofs. Exhibits a high capacity for scholarship and collaboration, going beyond core requirements to achieve learning goals. As a reference, the overall course score in A category (A-, A, A+) is no less than 90.00 (out of 100)
B	Good Performance	Shows good knowledge and understanding of the main subject matter of discrete mathematics, competence in problem-solving, and the ability of mathematical reasoning and proofs. Displays high motivation to learn and the ability to work effectively with others. As a reference, the overall course score in B category (B-, B, B+) is between 78.00 and 89.99 (out of 100).
C	Satisfactory Performance	Possesses adequate knowledge of core subject matter of discrete mathematics, competence in dealing with familiar problems, and some capacity for mathematical reasoning and proofs. Shows persistence and effort to achieve broadly defined learning goals. As a reference, the overall course score in C category (C-, C, C+) is between 66.00 and 77.99 (out of 100).
D	Marginal Pass	Has threshold knowledge of core subject matter of discrete mathematics, potential to achieve key professional skills, and the ability to make basic mathematical reasoning and proof. Benefits from the course and has the potential to develop in the discipline. As a reference, the overall course score in D category is between 60.00 and 65.99 (out of 100).
F	Fail	Demonstrates insufficient understanding of the subject matter of discrete mathematics and lacks the necessary problem-solving skills. Shows limited ability to mathematical reasoning and proofs, and exhibits minimal effort towards achieving learning goals. Does not meet the threshold requirements for professional practice or development in the discipline. As a reference, the overall course score in F category is below 60.00 (out of 100).

Late Submission Policy

- 1-3 Days Late: 10% of the total possible points will be deducted from the score.
- 4-7 Days Late: 20% of the total possible points will be deducted.
- More than 7 Days Late: The assignment will not be accepted and will receive a score of zero.
- Extensions:
 - Extensions will only be granted in exceptional circumstances, such as documented medical emergencies or other serious situations. Students must contact the instructor as soon as possible to request an extension.
 - If there are technical difficulties with the submission platform, students must email the instructor immediately with evidence of the issue (such as screenshots) and submit the assignment as soon as the issue is resolved.

Anything more?



What do we expect

- **Before attending each lecture:**
 - Read the textbook (around 30 pages)
- **During the lecture:**
 - Learn the concepts and examples (to enhance your understanding)
 - Work on quizzes (as a self-evaluation)
- **After the lecture:**
 - Work on the assignments independently (and review the textbook when necessary)
 - Work on the project (apply the knowledge and skills on real problems)
- **Tutorials by Teaching Assistants:**
 - More challenging examples and review of common mistakes in exercises
- **Office hours:**
 - Face-to-face discussions

Policy on the use of AI tools

- Students **are encouraged to** use generative artificial intelligence (GAI) tools for self-studying and exploring the topics of discrete mathematics.
- However, students are **not allowed** to use GAI tools to complete assessment tasks in this course.

Academic Integrity

● Academic integrity

- Students are expected to adhere to the university's academic integrity policy. Students are expected to uphold HKUST(GZ)'s Academic Honor Code and to maintain the highest standards of academic integrity. The University has zero tolerance of academic misconduct. Please refer to Regulations for Academic Integrity and Student Conduct for the University's definition of plagiarism and ways to avoid cheating and plagiarism.
- <https://vptlo.hkust-gz.edu.cn/ugeducation/#/ugstudenthandbook>

● How to avoid plagiarism and copying?

- <https://www.hkust-gz.edu.cn/academics/academic-quality-manual/roles-and-responsibilities-students/how-to-avoid-plagiarism-and-copying/>

● Example case:

- Student A borrowed the completed assignment from Student B for whatever reason. Student A then partially copied B's solutions and submitted his assignment.
- The software Turnitin detected that the submitted assignments from A and B have significant similarities. Course instructors investigated the case.
- Both A and B will receive zero marks on this assignment. The case will be reported to Thrust for further action.